



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

ELECTRICAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Electrical Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers and small rounding differences.

QUESTION 1

1 1.1 B ✓✓ 1.2 A ✓✓ 1.3 A ✓✓ 1.4 B ✓✓ 1.5 C ✓✓ [10]

QUESTION 2

- 2.1** An unsafe act is something a person does that can cause an accident ✓, e.g. working on a live circuit or not wearing safety glasses ✓. An unsafe condition is a dangerous state of the workplace or equipment ✓, e.g. a frayed cable, a wet floor or a missing machine guard ✓. (4)
- 2.2** A live circuit can cause electric shock, burns or death ✓. Any TWO steps: switch off and isolate the circuit at the distribution board ✓; lock the switch and attach a warning tag ✓; test with a tester that the circuit is dead before starting (3)
- 2.3** Any THREE: muscles contract so the person cannot let go ✓; burns at the entry and exit points ✓; the heart beats irregularly (fibrillation) or stops ✓; breathing stops; falls from a height (3)

[10]

QUESTION 3

- 3.1** (a) $V_{\max} = 340 \text{ V}$ ✓ (7)
- (b) $\omega = 2\pi f \rightarrow f = 314 \div 2\pi \checkmark = 50 \text{ Hz}$ (49,97 Hz) ✓
- (c) $V_{\text{rms}} = 0,707 \times 340 \checkmark = 240,38 \text{ V}$ ✓
- (d) $v = 340 \sin(314 \times 0,005) = 340 \sin(1,57 \text{ rad} = 90^\circ) \checkmark = 340 \text{ V}$ ✓
- 3.2.1** $X_L = 2\pi fL \checkmark = 2\pi \times 50 \times 0,2 \checkmark = 62,83 \Omega$ ✓ (3)
- 3.2.2** $X_C = 1 \div (2\pi fC) \checkmark = 1 \div (2\pi \times 50 \times 100 \times 10^{-6}) \checkmark = 31,83 \Omega$ ✓ (3)
- 3.2.3** $Z = \sqrt{R^2 + (X_L - X_C)^2} \checkmark = \sqrt{30^2 + 31^2} \checkmark = 43,14 \Omega$ ✓ (3)
- 3.2.4** $I = V \div Z \checkmark = 220 \div 43,14 \checkmark = 5,10 \text{ A}$ ✓ (3)
- 3.2.5** $\theta = \cos^{-1}(R \div Z) = \cos^{-1}(30 \div 43,14) \checkmark = 45,94^\circ$ ✓ (3)
- The current lags the voltage, because $X_L > X_C$ (the circuit is inductive) ✓
- 3.2.6** V_L and V_C are 180° out of phase and partly cancel each other ✓. The supply voltage is the phasor sum of V_R , V_L and V_C , so a single component voltage can be larger than the supply ✓. (2)
- 3.3** $f_T = 1 \div (2\pi\sqrt{LC}) \checkmark = 1 \div (2\pi\sqrt{0,2 \times 100 \times 10^{-6}}) \checkmark = 35,59 \text{ Hz}$ ✓ (3)

- 3.4 The impedance is a minimum and equals R ✓; the current is a maximum ✓; the power factor is 1 (current in phase with voltage) ✓. (3)

[30]

QUESTION 4

- 4.1 (a) P is connected to positive and N to negative ✓; the depletion layer narrows and current flows once about 0,7 V is exceeded ✓. (4)
(b) P is connected to negative and N to positive ✓; the depletion layer widens and only a tiny leakage current flows (until breakdown) ✓.
- 4.2 Positive half-cycle: one pair of diagonally opposite diodes (e.g. D1 and D3) conducts ✓ and current flows through the load in one direction ✓. Negative half-cycle: the other pair (D2 and D4) conducts ✓ and current still flows through the load in the same direction ✓, giving full-wave pulsating DC. The capacitor charges at the peaks and discharges into the load between them ✓, which smooths the output (reduces ripple) ✓. (6)
- 4.3 (a) $V_R = 15 - 9,1 \text{ ✓} = 5,9 \text{ V ✓}$ (8)
(b) $I_T = 5,9 \div 220 \text{ ✓} = 26,82 \text{ mA ✓}$
(c) $I_L = 9,1 \div 1\,000 \text{ ✓} = 9,1 \text{ mA ✓}$
(d) $I_Z = I_T - I_L = 26,82 - 9,1 \text{ ✓} = 17,72 \text{ mA ✓}$
- 4.4 $I_C = \beta \times I_B = 100 \times 40 \text{ } \mu\text{A ✓} = 4 \text{ mA ✓}$ (4)
 $I_E = I_C + I_B \text{ ✓} = 4,04 \text{ mA ✓}$
- 4.5 It limits the current so the LED is not damaged ✓ (3)
 $R = (12 - 2) \div 0,02 \text{ ✓} = 500 \text{ } \Omega \text{ ✓}$

[25]

QUESTION 5

- 5.1 $Y = A(B + \bar{B}) + \bar{A}B \text{ ✓}$ (5)
 $= A \cdot 1 + \bar{A}B \text{ ✓} = A + \bar{A}B \text{ ✓}$
 $= (A + \bar{A})(A + B) \text{ ✓} = A + B \text{ ✓}$

5.2

(6)

A	B	C	A + B	$\overline{C}$	Y
0	0	0	0	1	0
0	0	1	0	0	0
0	1	0	1	1	1
0	1	1	1	0	0
1	0	0	1	1	1
1	0	1	1	0	0
1	1	0	1	1	1
1	1	1	1	0	0

All 8 input combinations ✓; A + B column ✓; $\overline{C}$ column ✓; Y column ✓✓✓ (one mark for every two correct rows, rounded down)

5.3

(6)

	$\overline{B}C$	$\overline{B}\overline{C}$	BC	$B\overline{C}$
$\overline{A}$	0	1	1	0
A	0	1	1	1

Correct map ✓; group of four 1s (m1, m3, m5, m7) gives C ✓✓; pair (m6, m7) gives AB ✓✓

$$F = C + AB \quad \checkmark$$

5.4

(5)

The capacitor charges through R_1 and R_2 to $\frac{2}{3} V_{CC}$ while the output is high ✓. The comparator then resets the flip-flop, the output goes low and C discharges through R_2 to $\frac{1}{3} V_{CC}$ ✓. The cycle repeats on its own, giving a square wave ✓.

$$f = 1,44 \div ((10\,000 + 2 \times 47\,000) \times 10 \times 10^{-6}) = 1,44 \div 1,04 \checkmark = 1,38 \text{ Hz}$$

✓

5.5

(3)

In a combinational circuit the output depends only on the present inputs ✓. In a sequential circuit the output also depends on the previous state (it has memory) ✓, e.g. a flip-flop, counter or shift register ✓.

[25]

TOTAL: 100

END OF MARKING GUIDELINE